



UNIVERSIDAD DE DAGUPAN

SCHOOL OF INFORMATION TECHNOLOGY EDUCATION

ITP10 | EVENT-DRIVEN PROGRAMMING

PRELIM | LABORATORY ACTIVITY 2

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SCORE:

Debugging Exercise: Fixing a Basic Event-Driven Webpage

Objective

At the end of the exercise, students should be able to identify and correct common HTML and JavaScript errors, use the browser console to locate syntax and runtime errors, properly attach event listeners and handlers, and test whether an event-driven webpage functions correctly.

Activity Description

The following webpage contains several errors that prevent some of its features from working correctly. Use the browser Developer Tools and Console to identify and correct the errors.

The completed webpage should:

- Display a personalized greeting when the button is clicked.
- Show a message while the user is typing.
- Change the webpage background color.
- Reset the webpage to its original state.
- Display debugging messages in the browser console.
- Run without console errors.

Project Folder Structure

debugging-exercise/

```
|  
├── index.html  
├── style.css  
└── script.js
```

Part A: Buggy HTML Code

Save the following code as index.html.

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
  <meta charset="UTF-8">  
  <meta name="viewport" content="width=device-width, initial-scale=1.0">  
  
  <title>JavaScript Debugging Exercise</title>  
  
  <link rel="stylesheet" href="styles.css">  
  <script src="script.js"></script>  
</head>  
  
<body>  
  <main class="container">  
    <h1 id="mainHeading">Event-Driven Webpage</h1>  
  
    <label for="nameInput">Enter your name:</label>  
    <input
```

```

        type="text"
        id="nameInput"
        placeholder="Type your name"
    >

    <p id="typingMessage">You are typing:</p>

    <div class="button-group">
        <button id="greetButton">Display Greeting</button>
        <button id="colorButton">Change Background</button>
        <button id="resetButton">Reset</button>
    </div>

    <p id="resultMessage">
        Enter your name and select an action.
    </p>
</main>
</body>
</html>

```

Part B: CSS Code

Save the following code as `style.css`.

```

* {
    box-sizing: border-box;
}

body {
    margin: 0;
    min-height: 100vh;
    display: flex;
    justify-content: center;
    align-items: center;
    background-color: #f3f4f6;
    font-family: Arial, sans-serif;
}

.container {
    width: 90%;
    max-width: 600px;
    padding: 30px;
    text-align: center;
    background-color: white;
    border-radius: 10px;
    box-shadow: 0 4px 12px rgba(0, 0, 0, 0.15);
}

input {
    width: 100%;
    padding: 10px;
    margin-top: 8px;
}

.button-group {
    margin-top: 20px;
}

```

```
button {
    padding: 10px 15px;
    margin: 5px;
    cursor: pointer;
}

#resultMessage {
    padding: 12px;
    background-color: #f3f4f6;
}
```

Part C: Buggy JavaScript Code

Save the following code as script.js.

```
const heading = document.querySelector("#mainHeading");
const nameInput = document.querySelector("#name");
const typingMessage = document.querySelector("#typingMessage");
const resultMessage = document.querySelector("#resultMessage");

const greetButton = document.querySelector("#greetButton");
const colorButton = document.querySelector("#colorButton");
const resetButton = document.querySelector("#resetBtn");

function displayGreeting() {
    const userName = nameInput.value.trim();

    if (userName = "") {
        resultMessage.textContent =
            "Please enter your name.";
    } else {
        heading.textContent = "Hello, " + userName;
        resultMessage.textContent =
            "The greeting was displayed successfully."
    }
}

function showTypedText() {
    typingMessage.textContent =
        "You are typing: " + nameInput.value;
}

function changeBackground() {
    document.body.style.backgroundColor = lightblue;
    console.log("Background color changed.");
}

function resetPage() {
    heading.textContent = "Event-Driven Webpage";
    nameInput.value = "";
    typingMessage.textContent = "You are typing:";
    resultMessage.textContent =
        "Enter your name and select an action.";
    document.body.style.backgroundColor = "#f3f4f6";
}
```

```
greetButton.addEventListener("click", displayGreeting());
colorButton.addEventListener("clicked", changeBackground);
resetButton.addEventListener("click", resetPage);
nameInput.addEventListener("change", showTypedText);

console.log("JavaScript file loaded successfully.");
```

Instructions

1. Create the required project folder and files.
2. Copy the provided code into the correct files.
3. Open `index.html` in a web browser.
4. Open the browser Developer Tools and select the **Console** panel.
5. Identify the errors shown in the console.
6. Correct all errors in the HTML and JavaScript files.
7. Test every required feature after making corrections.
8. Record each error in the debugging table.
9. Submit the corrected files and screenshots of the working webpage.
- 10.

Debugging Table

No.	Error Found	Type of Error	Correction Made
1	styles.css:1 Failed to load resource: net::ERR_FILE_NOT_FOUND <code><link rel="stylesheet" href="style.css"></code>	Incorrect file reference	Renamed the stylesheet reference "styles.css" to "style.css" in HTML to match
2	script.js:40 Uncaught TypeError: Cannot read properties of null (reading 'addEventListener') <code>const resetButton = document.querySelector("#resetBtn");</code>	Incorrect element selector	Updated the query selector in script.js from <code>#resetBtn</code> to <code>#resetButton</code> to match the HTML IDs.
3	script.js:40 Uncaught TypeError: Cannot read properties of null (reading 'addEventListener') <code>const nameInput = document.querySelector("#name");</code>	Incorrect element selector	Updated the query selector in script.js from <code>#name</code> to <code>#nameInput</code> to match the HTML IDs.
4	<code>if (userName = "")</code>	Logical error	Changed the assignment operator <code>=</code> to strict comparison <code>===</code>
5	<code>colorButton.addEventListener("clicked", changeBackground);</code>	Incorrect event type	Changed the invalid event name "clicked" to "click".
6	<code>greetButton.addEventListener("click", displayGreeting());</code>	Incorrect event-handler assignment	removed the parenthesis in <code>displaygreeting</code> so the function is referenced instead of execute

7	<pre>nameInput.addEventListener("input", showTypedText);</pre>	Incorrect event type	changed "change" to "input" so the message updates live while typing
8	<pre>document.body.style.backgroundColor = lightblue;</pre>	Runtime error	put lightblue inside quotation mark
9	script.js:41 Uncaught TypeError: Cannot read properties of null (reading 'addEventListener')	Runtime error	moved the script tag <code><script src="script.js"></script></code> to the bottom of the body
10	index.html:1 Unsafe attempt to load URL file:///C:/Users/User/Desktop/debugging-exercise/index.html from frame with URL file:///C:/Users/User/Desktop/debugging-exercise/index.html. 'file:' URLs are treated as unique security origins.	Runtime error	Ran the index.html using the Live Server Extension

Possible error types include:

- Syntax error
- Runtime error
- Logical error
- Incorrect file reference
- Incorrect element selector
- Incorrect event type
- Incorrect event-handler assignment

Testing Requirements

After correcting the program, verify that:

- The CSS file is properly linked.
- The JavaScript runs after the HTML elements are available.
- Typing in the input field immediately updates the typing message.
- Clicking **Display Greeting** shows the entered name.
- Empty input displays a warning message.
- Clicking **Change Background** changes the page color.
- Clicking **Reset** restores the original page.
- All buttons work using `addEventListener()`.
- The console displays debugging messages.
- No errors appear in the browser console.

Submission Requirements

- Submit the corrected project folder containing:
 - index.html
 - style.css
 - script.js
- Include the completed debugging table.
- Submit screenshots showing:
 - The original console errors
 - The corrected webpage
 - The working personalized greeting
 - The changed background color
 - The browser console without errors
- Compress the files into one ZIP folder.
- Use the filename format:

Surname_FirstName_DebuggingExercise.zip

Performance Checklist

Performance Indicator	Performed Correctly	Needs Improvement	Points
Created the required project folder and files.	☐	☐	5
Corrected the CSS filename or file reference.	☐	☐	5
Ensured that JavaScript runs after the HTML elements load.	☐	☐	5
Corrected all incorrect element selectors or IDs.	☐	☐	10
Corrected the conditional statement used for empty input.	☐	☐	10
Corrected the background-color value.	☐	☐	10
Properly passed handler functions to <code>addEventListener()</code> .	☐	☐	10
Corrected the invalid event type.	☐	☐	10
Used the appropriate event to display text while typing.	☐	☐	10
Greeting and empty-input validation work correctly.	☐	☐	5
Background-change and reset features work correctly.	☐	☐	5
Used the browser console to identify and verify errors.	☐	☐	5
Completed the debugging table accurately.	☐	☐	5
Submitted complete files and required screenshots.	☐	☐	5
Total			100

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